

# Kaustubh Rathi

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## SUMMARY

Aspiring Quantitative Researcher with a strong mathematical foundation in stochastic calculus, probability, and statistical modeling, and self-directed research across derivative pricing and stochastic processes. Competitive programmer (Codeforces Expert, LeetCode Knight) with production software engineering at Amazon; fluent in Python (NumPy/SciPy) and C++. Minor in Economics (econometrics).

## EDUCATION

### B.Tech in Information Technology (Minor in Economics)

2022 – 2026

Indian Institute of Information Technology (IIIT), Allahabad — CGPA: 8.52/10

Prayagraj, India

**Relevant Coursework:** Probability & Statistics, Linear Algebra, Game Theory, Data Mining, Machine Learning, Econometrics (Economics minor).

## TECHNICAL SKILLS

**Mathematics & Statistics:** Stochastic Calculus, Probability, Statistics, Linear Algebra, Optimization, Time-Series Analysis, Monte Carlo Methods, Numerical Methods

**Quantitative Finance:** Derivative Pricing, Black-Scholes, Option Greeks, Stochastic Processes (Vasicek, CIR, CEV, Bessel), American-Option Pricing (Longstaff-Schwartz), Malliavin Calculus

**Programming:** Python (NumPy, SciPy, Pandas), C++, Java, SQL

**Tools:** Linux, Git, Docker, Matplotlib, statsmodels

## QUANTITATIVE RESEARCH & PROJECTS

### Quantitative Finance Research — Stochastic Processes & Derivative Pricing | Python, NumPy, SciPy

🐙 GitHub

- Implemented and analyzed 6 continuous-time stochastic-process simulators (Brownian motion with drift, GBM, Vasicek, CIR, CEV, Bessel) over 10,000-path ensembles, studying path distributions, parameter sensitivity, and mean-reversion behavior for interest-rate and asset-price modeling.
- Priced options via 4 methods from first principles: closed-form Black-Scholes, 100k-path Monte Carlo, Longstaff-Schwartz least-squares for American options, and Malliavin-calculus Greeks estimation; verified MC prices converge to within 0.5% of analytical Black-Scholes.

### Black-Scholes Option Pricer | Python, Streamlit, py\_vollib, SciPy

🐙 GitHub

- Deployed a Streamlit tool that fetches live market prices and computes European option prices and the full Greeks (delta, gamma, vega, theta) with dividend-yield adjustment.

### Neural Collaborative Filtering | PyTorch Lightning, MovieLens

🐙 GitHub

- Trained a matrix-factorization recommender on MovieLens-1M (1M+ ratings), learning 50-dim latent user/item factors and reaching 0.79 MAE on an 80/20 hold-out — ~16% below a global-mean baseline (0.94); applied statistical modeling end to end.

## EXPERIENCE

### Software Development Engineer Intern

Jan 2026 – Present

Amazon — Grocery Ordering, Stores

Bengaluru, India

- Developed statistical data-correlation tooling over ~150 daily order-failure events, joining multi-source telemetry (Cloud-Watch, Step Functions, deployment history) to automatically cluster and attribute failures to root causes.
- Engineered an autonomous remediation pipeline (deterministic state machine, output validation, retries, 1–5 confidence scoring, human approval gates) and the Java backend (Lambda + API Gateway, DynamoDB); reduced incident investigation time from ~2 hours to ~15 minutes.

### Artificial Intelligence Intern

Dec 2024 – Feb 2025

ASPR Agrovalue (I) Limited

India

- Automated a Python web-scraping + ETL pipeline aggregating 1,000+ daily market/price records and an NER invoice-extraction pipeline (10,000+ invoices/month), feeding a Streamlit analytics dashboard adopted by the procurement team.

## ACHIEVEMENTS

- Codeforces Expert** (max rating 1751) | **LeetCode Knight** (rating 2000+), 800+ algorithmic problems solved.
- Secured **99.45 percentile** in JEE Mains 2022 (top 0.55% among 1M+ candidates nationwide).
- Completed *Options 101* by Akuna Capital — derivatives and options-trading fundamentals.